Option Characteristics and Volatility Smirks as Predictors of Equity Returns

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AGENDA

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Motivation

Why Look at Option-Implied Signals?

[1] **Equity options reflect how investors view risk, uncertainty, and potential mispricing. The shape of the implied volatility curve often reveals information that stock returns alone do not capture. Prior research shows that features related to the volatility smirk can forecast returns, and our goal is to examine whether these signals still matter in more recent market conditions. Understanding this connection can improve both asset pricing models and systematic trading decisions.**

[2] **Key Ideas**

- Do options have information not in returns
- Does this hold in today's market conditions
- Can this improve models

Research Questions

A	**Predictive Power**	Which option-implied signals, including the volatility smirk, skew measures, and related features, show meaningful ability to forecast cross-sectional stock returns?
B	**Systematic Strategies**	Can these signals be turned into simple, rules-based strategies that add value beyond standard equity factors?
C	**Economic Interpretation**	What do these smirk-related patterns tell us about investor behavior, mispricing around earnings, and how quickly information gets reflected in prices?

Option Characteristics and Volatility Smirks as Predictors of Equity Returns

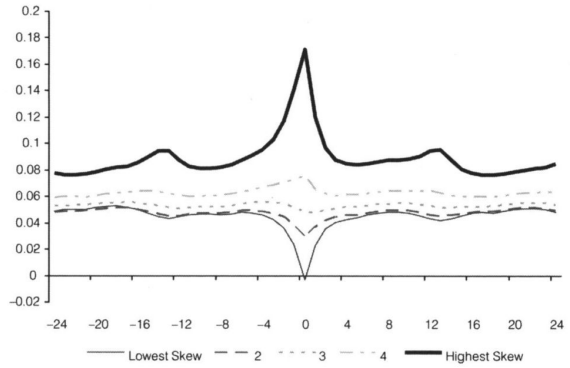
Academic Literature

What Does the Individual Option Volatility Smirk Tell Us About Future Equity Returns?

FIGURE 1

Evolution of Volatility Skew over [–24, +24]

Data are obtained from CRSP, Compustat (for stocks), and OptionMetrics (for options). Our sample period is 1996–2005. Variable SKEW is the difference between the implied volatilities of out-of-the-money (OTM) put options (strike-to-stock-price ratio closest to 0.95) and at-the-money (ATM) call options (strike-to-stock-price ratio closest to 1). In Figure 1, we track the average volatility skew for firms within quintile portfolios between 24 weeks before the sorting and 24 weeks after the sorting, while the ranks of quintile portfolios are determined based on SKEW at week 0.



The figure shows how volatility skew varies across portfolios sorted by the skew measure. Firms with the highest skew display a persistent elevation in implied volatility for out of the money puts, which reflects strong demand for protection.

$$\mathrm{SKEW}_{i,t} = \mathrm{VOL}_{i,t}^{\mathrm{OTMP}} - \mathrm{VOL}_{i,t}^{\mathrm{ATMC}}.$$

This measure captures how much more investors are willing to pay for downside protection. A larger gap between put and call implied volatilities signals stronger demand for insurance against negative moves.

$$\mathrm{RET}_{i,t} = b_{0t} + b_{1t}\mathrm{SKEW}_{i,t-1} + b_{2t}'\mathrm{CONTROLS}_{i,t-1} + e_{it},$$

The authors regress future stock returns on the skew measure while controlling for standard firm characteristics. They find that stocks with higher skew tend to earn lower returns in the following period.

This paper provides the foundation for our study, since it links the shape of the volatility smirk to future returns. We build on this idea by testing whether similar signals remain informative in more recent markets and around earnings events.

Option Characteristics as Cross-Sectional Predictors

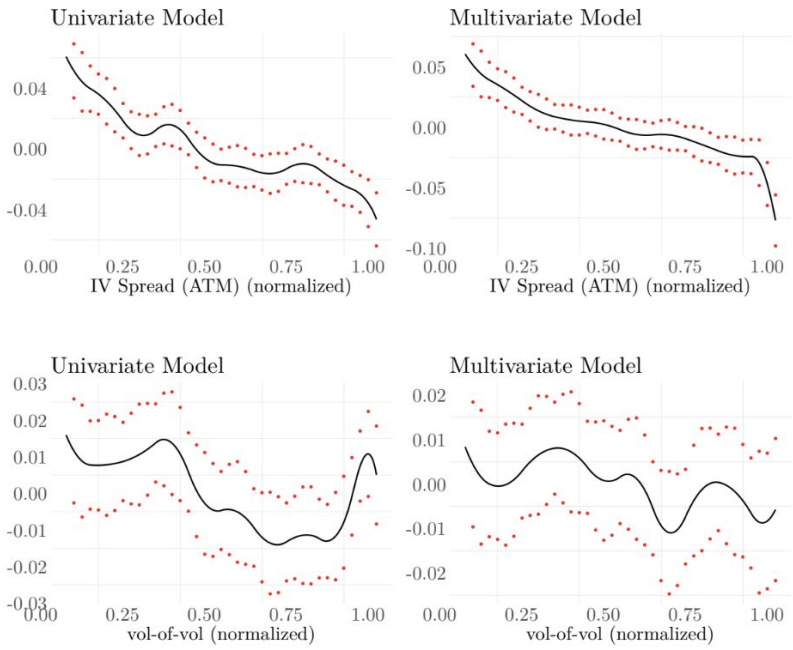
(Neuhierl, Tang, Varneskov, and Zhou in 2022)

The authors study a broad set of option characteristics to see which ones truly predict stock returns when they are evaluated together. Their goal is to move beyond single variable tests and understand which signals survive in a more realistic, joint framework.

They find that many predictors lose their strength once additional controls are included, but smirk related measures such as the at the money IV spread remain among the most reliable. The shape of the smirk carries information that does not disappear when other characteristics are added.

The figure highlights this point by showing how higher IV spreads are linked to lower future returns in both simple and multivariate settings. This result supports our focus on smirk based features and motivates testing whether these patterns still hold in more recent markets.

Figure 3: **Conditional Mean Functions - Selected vs. Not Selected Predictors**



The upper panel of the figure shows the conditional mean function for the IV spread (ATM), one of the strongest predictors, in a univariate model (left) and a multivariate model (right). The multivariate model includes the variable itself as well as the additional predictors selected by the adaptive group LASSO. A complete overview the variables is provided in Table 5. The lower panel provides the corresponding plots for the volatility-of-volatility variable. Specifically, "Vol. of vol" shows some predictive ability in the univariate model, but is not selected by the adaptive group LASSO in the multivariate model. The red dots indicate pointwise 95% confidence bands.

Why Do Option Prices Predict Stock Returns? Evidence from Analyst Tipping

(Lin and Lu in 2014)

$$spread_{i,t} = \sum_{j=1}^{n_{i,t}} w_{j,t}^{i}(IV_{j,t}^{i,call} - IV_{j,t}^{i,put}),$$

$$skew_{i,t} = IV_{i,t}^{OTMP} - IV_{i,t}^{ATMC},$$

This paper examines whether simple option-implied measures can explain differences in stock returns. The authors show that characteristics like call put spreads and volatility skew contain information that is not reflected in stock prices alone. The results support our focus on smirk-related measures and guide the way we build and test our predictors.

Table 5
IV spread/skew and earnings performance.

		IV proxy quintiles			
		High	Median	Low	High–Low
Panel A: IV spread					
Analyst initiation	Preceding SUE	0.039%	0.059%	0.045%	−0.006%
	t-stat	(5.15)	(9.30)	(4.78)	(−0.53)
	N	10,503	10,504	10,503	
Panel B: IV skew					
Analyst initiation	Preceding SUE	0.054%	0.072%	0.063%	−0.009%
	t-stat	(5.71)	(10.50)	(10.53)	(−0.80)
	N	7163	7164	7164	

This table displays the mean of earnings performance for sub-samples categorized by the IV spread and the IV skew. For each analyst's initial recommendation, stocks are sorted into terciles based on the average IV spread or IV skew over three days prior to the events. High refers to the top tercile with the highest IV spread/skew, and Low is the bottom tercile with the lowest IV spread/skew. SUE is the standardized unexpected earnings, which is calculated as the announced quarterly EPS less the corresponding consensus analyst forecast and scaled by the standard deviation of the quarterly earnings estimates. Preceding SUE is the mean of the SUEs in the quarter that precedes the analyst initiation. N is the number of events for each tercile. The column High–Low reports the difference in the mean of Preceding SUEs between high and low IV spread/skew sub-samples.

Gaps in Literature

THE TIMELINE

Much of the evidence linking smirks to future returns comes from periods with very different options markets.

Retail activity, weekly options, and zero day contracts expanded after 2020, which may change how option signals behave.

2010
Xing et al. show smirk predicts returns

2010
Johnson & So show informed trading patterns

2020
New evidence on analyst tipping

2020
Explosion in retail options activity and weekly/zero-day options

2025
Our Work

Data Overview

Data Sources and Structure

Crisis Datasets (SPX Options)

spx2008.csv
spx2020.csv

- Measure smirk during 2008 & 2020 crashes
- Capture volatility spikes and downside insurance demand

Model Datasets (Equity Options)

data2008.csv
data2020.csv

- Filtered versions used for feature construction across equities
- Ensures consistent strike/maturity structure

Modern Dataset (Equity Options)

alloptions.csv

- Around 230k rows
- Full cross-section of equity options
- Used to test whether smirk still predicts returns

Table Structure

(Shared Across Datasets)

- Date
- Expiration
- Strike
- Call/put
- Bid
- Ask
- Mid volume
- Open Interest
- Implied Volatility
- Underlying Price

Derived Fields Used

- iv_call
- iv_put
- smirk_iv = iv_put – iv_call
- smirk_price = mid_put – mid_call
- log_moneyness
- 1–5 day future returns

Data Selection

Crisis Periods (2008 & 2020)

- Provide two structurally different stress environments
- Allow testing whether smirk predictive power strengthens in crises
- Benchmark against academic literature that often uses pre-2012 data

Modern Cross-Section (2025)

- The modern options market is dominated by:
  - 0DTE
  - retail-driven flow
  - tighter spreads
  - faster IV reactions
- Older smirk studies may no longer hold
- The alloptions.csv provides a clean test of smirk in 2025 microstructure

Ticker	Date (MM/DD)	Direction
DDOG	10/31	+23.08%
NET	11/07	+16.04%
AMD	10/29	+7.42%
QCOM	11/06	+6.34%
MU	11/27	+5.77%
AOS	05/28	−6.30%
PYPL	11/01	−6.89%
EL	11/01	−18.91%
CMCSA	10/30	−6.12%
SWKS	11/06	−7.53%
AAPL	11/02	+1.23%
KO	10/24	+2.32%
PEP	10/10	+1.98%
CMG	10/24	+3.09%
VZ	10/24	+3.52%
GOOGL	10/26	−0.96%
META	10/25	−1.02%
UNH	10/13	−1.25%
JNJ	10/17	−0.56%
NFLX	10/18	−0.20%

Methodology & Models

Modeling Pipeline

Data Inputs

- Crisis datasets: 2008, 2020 SPX options
- Modern cross-section: 2025 equity options
- Earnings event sample: 20 tickers with known reaction direction

Feature Engineering

- Mid prices, implied volatilities (solver)
- Smirk metrics
  - IV smirk
  - Price smirk
- Log moneyness & days to expiration
- Liquidity variables (bid-ask, volume, OI)
- Expiration days

Target Variables

- Next-day stock return (regression)
- Next-day direction (classification)

Modeling Design

- Time-series forward split
- Regime-based comparisons
- Crisis vs modern vs earnings events

Feature Construction

Why it Matters

Captures demand for downside protection

Robust across crises and modern markets

Controls isolate smirk's incremental effect

Aligns with Xing et al. (2010)

Dataset Preview

symbol	date	mid_call	mid_pu	iv_call	smirk_iv	smirk_price	log_moneyness	exp_dom	oi	target_return
AMZN	2008-12-23	0,781	0,684	0,858	0,505	0,252	0,26	1085		0,03
GOOG	2008-12-22	0,758	0,780	0,839	−0,201	−0,056	−0,03	2.826		−3,08
MSFT	2008-12-29	0,782	0,704	0,713	−0,167	0,33	0,28	2.023		−11,50
SPY	2008-12-29	0,526	0,673	0,487	0,065	0,125	0,22	19.060		0,35
META	2008-12-28	0,703	0,674	0,678	−0,117	0,136	0,20	218		0,27

IV Smirk:

$$\mathrm{IV}_{\mathrm{put}} - \mathrm{IV}_{\mathrm{call}}$$

Price Smirk:

$$\mathrm{mid}_{\mathrm{put}} - \mathrm{mid}_{\mathrm{call}}$$

Log Moneyness:

$$\log(S/K)$$

Controls:

bid–ask spread, volume, OI, expiration days

Earnings Event Modeling Framework

Pre-Earnings Window

For each stock, build a 3–5 day window before the announcement

Pull all option chains from each “window day”

Compute smirk features (IV smirk, price smirk) daily

Event Labeling

Compute next-day return after earnings

Convert to direction labels:

positive, negative, or neutral

Final Earnings

20 tickers with clear earnings reactions

Includes:

ticker, earnings date, smirk measures, reaction direction

Purpose of Earnings Testing

Earnings are high-information events with asymmetric expectations

Traders hedge aggressively before announcements → smirk reacts early

Clean way to test whether smirk predicts short-horizon returns

Mirrors methodology from Xing et al. (2010)

What We Learn

Whether smirk increases before negative surprises (downside hedging)

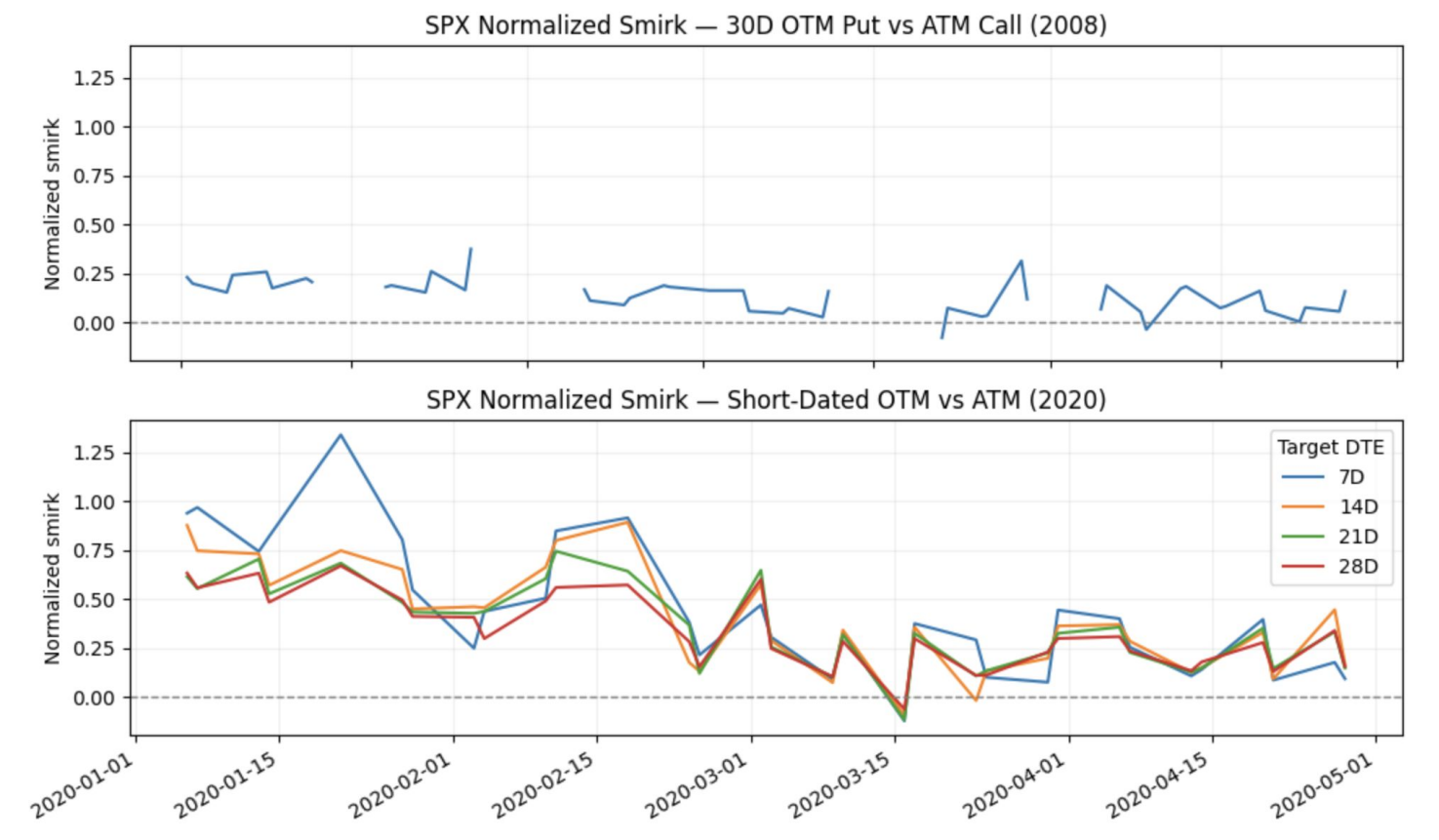
Whether smirk flattens before positive surprises

If smirk retains predictive power in modern markets

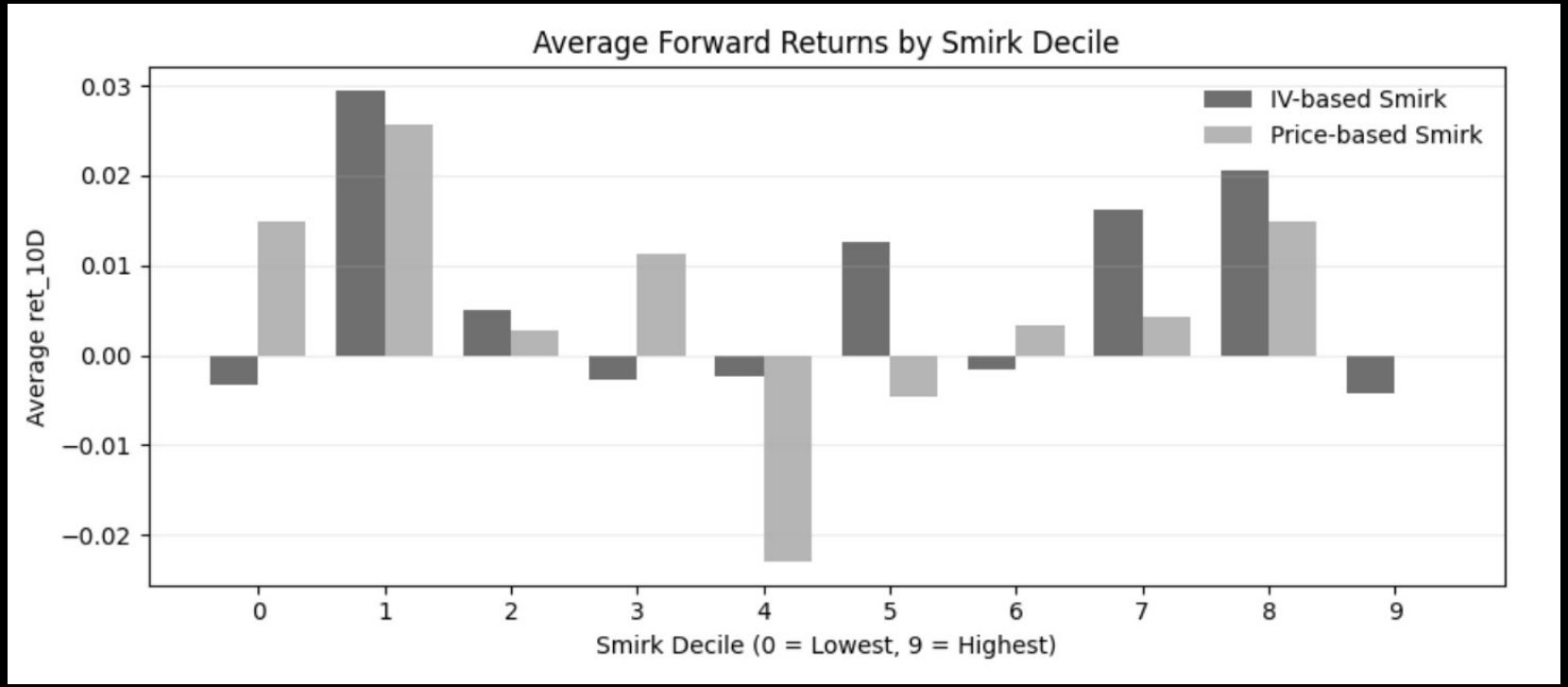
If the pattern differs from crisis periods

Results & Conclusion

Crisis Period Response of Smirk

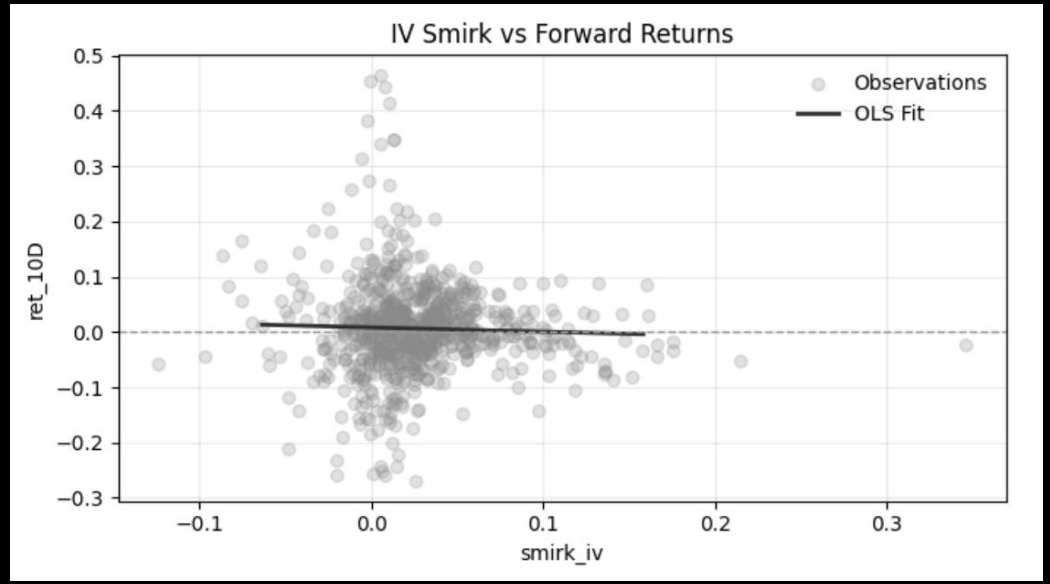


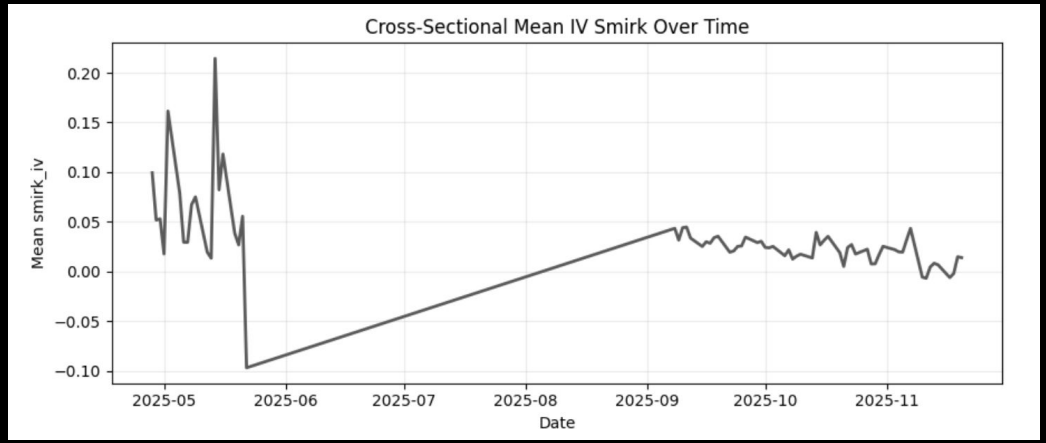
Modern Smirk Signals: Decile Evidence



Higher smirk levels are associated with higher 10-day forward stock returns in the modern cross-section. The IV-based smirk exhibits a clearer monotonic pattern, with higher deciles earning stronger average returns. The price-based smirk shows a similar but noisier relationship. Overall, smirk retains predictive power in modern markets, though the signal is weaker than during crisis periods.

Modern Smirk Signals: Time-Series & Scatter





Positive but weak relationship:
IV smirk is associated with higher short-horizon returns

Low explanatory power:
High noise consistent with efficient modern markets

Contrast with crises:
Signal is materially weaker than during 2008 and 2020

Smirk Predictability Across Market Regimes

Regime	Evidence	Strength	Interpretation
2008 / 2020 Crisis	Time-series spikes (R1)	**Strong**	Demand for downside protection is acute
Modern Cross-Section	Decile & scatter (R2–R3)	**Moderate / Weak**	Signal diluted by efficiency
Earnings Events	Pre-event smirk sorting (R3)	**Localized**	Informed trading concentrates signal

Key Findings

- **Smirk predictability is strongly regime-dependent**
  Option-implied smirk signals are most informative during periods of market stress and around discrete events, rather than uniformly across time.
- **Crisis periods exhibit the strongest smirk response**
  During 2008 and 2020, smirk measures spike sharply, reflecting acute demand for downside protection and heightened risk aversion.
- **In modern markets, smirk remains predictive but is weaker and noisier**
  Cross-sectional decile tests and scatter evidence show a positive relationship with future returns, though the signal is diluted by increased market efficiency.
- **Earnings events concentrate smirk information locally**
  Pre-earnings smirk patterns align with next-day reaction direction, suggesting informed trading temporarily restores predictive power.

THANK YOU!

Any Questions?

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